Phase 3 — External Data (yFinance)

fetcher · indicators · cache · ticker_map · market_hours · models · timeutil · exceptions — 100% code, no LLM

● phase 3 · live-verified

input

IG epic
IX.D.DAX.IFMM.IP

external_data/ — MarketDataFetcher pipeline — single network point ↓⑤

public API · MarketDataFetcher(mapper, cache, rate_limiter)

- intraday → None off-hours: get_drift · get_momentum · get_bars
- daily (always): get_history · get_volume_anomaly · get_high_low_distance
- composite: get_brain_context() → BrainContext (fail-tolerant)

① ticker_map · epic → ^GDAXI
+ optional vol-proxy EXS1.DE

② market_hours.is_market_open()
Mo-Fr 09:00-17:30 Europe/Berlin (DST)

③ FileCache · JSON + TTL (atomic)
1d→midnight · 5m→120s · 30m→300s

④ RateLimiter · ≥ 0.5 s between calls

⑤ _raw_download() ← ONLY net call
retry: exp-backoff 1→2→4s · ≤3 tries

yfinance ≥0.2.65 (curl_cffi internal)
no custom session passed

⑥ indicators.py · pure fns
drift% · momentum(5m) · vol z-score · high/low

Yahoo Finance · ^GDAXI
delayed quote ~15 min · index = 0 volume
→ ETF proxy EXS1.DE for real volume

vol anomaly z thresholds ±2 / ±3 · high/low band 90d · near < 1.5% · daily history ffill + IQR-outlier

models.py · BrainContext
.to_prompt_dict() → LLM · .to_dict() → market_context_json

consumers (later phases) — cross-phase via public contracts only

Phase 4 · research | Phase 5 · gates (context only ⚠) | Phase 7 · longterm brain

contracts · models.py (ISO-UTC, ms, Z)

- PriceBar ohlcv · volume 0.0 if none
- DriftResult drift_pct · is_realtime=False
- MomentumResult 5m · is_realtime=False
- VolumeAnomaly z · volume_available · source
- HighLowDistance dist hi/lo · near <1.5%
- BrainContext composite · partial-ok

exceptions: ExternalDataError → EpicNotMapped · DataUnavailable · RateLimit (own tree, isolated)

key design decisions

- intraday off-hours → None (never stale)
- momentum is_realtime=False → context only
- ^GDAXI no volume → degrade gracefully
  + register_volume_proxy(epic, "EXS1.DE")
- dual error modes, both retried:
  YFRateLimitError → RateLimitError
  empty/short df → DataUnavailableError
- one network point: _raw_download()
- phase isolation: own timeutil, no broker_wrapper / persistence imports

⚠ Phase-5 carry-over (cross-phase notice)

hard momentum VETO must source from IG real-time bars (get_ohlcv) , NOT get_momentum() — yFinance is ~15 min delayed. Phase-3 momentum = context signal.

● live-verified · 2026-06-02

70 mocked tests green · pytest no-network
live_test 6/6 · ETF proxy EXS1.DE z=-1.36

build order: exceptions/models → timeutil/market_hours → ticke
10 steps · standalone · single network site · consumed by Phas